Session 5: Solutions and facilitation notes

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Exercise numbers follow `Practicals/s5_exercises.md`. All FLUID-ICU numbers come from the results packs in `Course/Data/results_packs/` (simulated data).

Exercise 1: Classify the variables

#	Variable	Answer	Explanation
a	Age (years)	**Continuous**	Measured on a scale. Keep the exact number on the CRF (or collect date of birth).
b	Sex	**Binary**	Two categories. Coding 1/2 does not make it numerical.
c	Glasgow Coma Scale	**Ordinal**	Ordered scores (3–15) with unequal steps. Often *analysed* like a number, which is acceptable if stated, but it is ordinal by nature.
d	Source of sepsis	**Nominal**	Several categories, no natural order.
e	Number of vasopressors	**Count**	Whole numbers 0–3. Accept "ordinal" if the participant argues it well. It is not continuous.
f	Cumulative fluid balance (ml/kg)	**Continuous**	Can be negative or positive, with any value on the scale. The Paper 1 primary outcome.
g	Alive at day 28	**Binary**	Yes/no.
h	Time to death, censored at day 28	**Time-to-event**	Two pieces of information: how long, and whether the event happened. Analysed with Kaplan–Meier / Cox.
i	Pain none–severe	**Ordinal**	Ordered categories.
j	Blood group	**Nominal**	No order.

Bonus: any continuous variable (age, fluid balance, lactate). You can always group later, but you cannot recover the exact value from a group.

Instructor note: common errors are calling GCS, time to death (this ignores censoring) or the number of vasopressors continuous.

Exercise 2: Fix the bad CRF items (model answers)

#	Bad item	Problem	Model rewrite
a	Fluid given: ____	No unit, no time window, no definition (crystalloid? blood? drug diluents?)	"Total IV fluid received from ICU admission to 24 h (ml), including crystalloids, colloids and drug diluents: [ ][ ][ ][ ][ ] ml □ not recorded"
b	Weight: ____	No unit. Measured or estimated? When?	"Body weight at ICU admission (kg): [ ][ ][ ].[ ] □ measured □ estimated □ not available"
c	Comorbidities: _____	Free text: impossible to analyse, spelling varies	Tick list: "Comorbidities present at admission (tick ALL that apply): □ diabetes □ chronic kidney disease □ heart failure □ COPD □ cirrhosis □ cancer □ none □ unknown"
d	Lactate high? Y / N	Opinion ("high" by whose cut-off?), loses the value, no time point	"Serum lactate at ICU admission (mmol/L): [ ][ ].[ ] □ not done" (define "high" later, in the analysis)

Instructor note: accept any version that passes the four checks (unit, time point, one reading, unknown option). Praise answers that notice that collecting the *number* is better than collecting a yes/no.

Exercise 3: Choose the right summary

#	Variable	Shape / type	Summary	Graph
a	Age, bell-shaped	Symmetric continuous	**Mean (SD)**	Histogram
b	ICU length of stay	Right-skewed	**Median (IQR)**	Histogram or boxplot
c	Died in hospital	Binary	**n (%)**	Bar chart
d	CRP	Right-skewed	**Median (IQR)**	Histogram / boxplot
e	Source of sepsis	Nominal	**n (%) per category**	Bar chart (not 3-D pie)
f	SBP by treatment group	Symmetric, two groups	**Mean (SD) per group**	Side-by-side boxplots

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Exercise 4: Judge a bad figure

1. Each bar is the **mean** fluid balance per group. The error bar is the **standard error (SE)**, shown in one direction only. Missing: the individual patients, the spread (SD or IQR), the shape (skew), the n per group on the figure, and what the error bar represents.

2. Any two of: the data are **skewed**, **one extreme patient** (160 ml/kg) pulls the usual-care mean up, the groups **overlap** a lot, the median, the number of patients, negative values.

3. Dots (one per patient) plus a boxplot per group (see the "dots + box" slide), y-axis labelled with units, n under each group. You gain: shape, spread, outliers and overlap become visible, and the reader can judge the comparison.

4. The truncated axis makes a 2-percentage-point difference (31% vs 29%) look like a three-fold difference. Fix: start bars at 0 and label the values (or show the difference with its 95% CI).

5. (i) The author chose the two scales, so the lines can be made to cross or move together wherever wanted. (ii) Two trends over time moving together is **correlation, not causation** (other changes may have happened at the same time). Better: two panels sharing the time axis, and an appropriate design (e.g. interrupted time series, Session 3).

Exercise 5: [FLUID-ICU] Spot the problems in the raw FLUID-ICU data

(See `Course/Data/cleaning_log.md` for the full log.)

#	What is wrong	What the script does
a	One variable coded many ways (and a leading space)	Recode to `Female` / `Male`
b	Dates in different formats (03/04: March or April?)	Read each format explicitly and convert to one date format (YYYY-MM-DD)
c	ICU C reported lactate in mg/dL, not mmol/L (about 9 times larger, found by comparing medians by ICU)	Divide ICU C values by 9.01 (1 mmol/L = 9.01 mg/dL), and confirm with the site laboratory
d	`999` is a missing-value code, not a lactate	Set to missing (NA)
e	Impossible age (outside the allowed range)	Set to missing, send a query to the site, and keep the patient
f	Duplicate record	Check the two rows are identical, keep one copy (623 rows become 620 patients)

Bonus: editing by hand is invisible and not reproducible. A script plus a log lets anyone check and repeat every change. The log records the problem, how many records, what was done, and why (plus queries sent).

Exercise 6: [FLUID-ICU] Read the FLUID-ICU Table 1

All numbers come from Table 1 of the Q1 results pack (the same table is in the Q2 and Q3 packs).

1. Negative balance n = 245, positive n = 373, overall 618 (column headings of Table 1). Two of the 620 enrolled patients have no day-3 fluid balance (weight or output missing after cleaning), so they cannot be placed in a group (participant flow, section 2).

2. Model answers:

Variable	Summary used	Why this summary?
Age	Mean (SD): 64.6 (15.0) vs 61.4 (15.2) years	Roughly symmetric

Variable	Summary used	Why this summary?
SOFA score	Median (Q1, Q3): 6 (4, 8) vs 8 (6, 10)	A skewed score
Vasopressor on admission	n (%): 61 (25%) vs 205 (55%)	Binary (yes/no)

3. No. The positive-balance group was sicker. The biggest difference is vasopressor on admission: 205 (55%) vs 61 (25%). SOFA (8 vs 6) and mechanical ventilation (216 (58%) vs 97 (40%)) point the same way.

4. Lactate, with 91 missing.

5. Model sentences: "Of 620 enrolled patients, 618 had a day-3 fluid balance, and 373 of them had a positive balance. Patients with a positive balance were more severely ill at admission (median SOFA 8 vs 6, vasopressor 55% vs 25%, mechanical ventilation 58% vs 40%)."

6. Larger. The positive-balance group was sicker, and sicker patients die more, so the crude association mixes fluid with severity (confounding by indication).

Instructor note: heart failure goes the other way, 43 (12%) vs 54 (22%), because clinicians give less fluid in heart failure. Session 6 confirms question 6: crude OR 2.49 vs adjusted 1.62 (Q1 pack, Table 3).

Exercise 7: [FLUID-ICU] A FLUID-ICU question as a PECO sentence

In class, everyone fills in the table for Q1 alone, then a few people read their sentence. Participants fill it in again after choosing their own question at the end of Session 6.

The table for Q1

	Model answer
The question (Q1 to Q5), copied from page 1 of its results pack	Q1: In adults with sepsis in the ICU, is a positive cumulative fluid balance by day 3 associated with higher 28-day mortality?
P: population	Adults with sepsis admitted to one of the four ICUs
E: exposure (variable name in the data dictionary) and its type	Positive fluid balance by day 3, `positive_balance_d3` (binary). It is derived from `cum_fluid_balance_d3_ml_kg` (continuous)
C: comparator	A negative or zero fluid balance by day 3
O: outcome (variable name) and its type	Death from any cause by day 28, `died_28d` (binary). Also time to death, `time_to_event_28d` (time-to-event)
One-sentence research question	"In adults with sepsis admitted to the ICU, is a positive cumulative fluid balance by day 3, compared with a negative balance, associated with higher 28-day mortality?"
Which test will compare the groups? Which model will adjust?	Chi-square test for the crude comparison. Logistic regression to adjust. Kaplan-Meier and Cox regression for time to death

When you redo the table for your own question (Q2 to Q5)

Q	E: exposure (type)	O: outcome (type)	Test, then model
Q2	Cumulative balance, `cum_fluid_balance_d3_ml_kg` (continuous, per 10 ml/kg)	AKI, KDIGO stage 1 or more, `aki_any` (binary)	Chi-square by group, then logistic regression
Q3	Cumulative balance (continuous)	Ventilator-free days to day 28, `vent_free_days_28` (count 0 to 28, skewed, piled at 0 and 28)	Wilcoxon rank-sum (Mann-Whitney U) test, then linear regression
Q4	Lactate at admission, `lactate_mmol_l`, mmol/L (continuous, 15% missing)	Death by day 28, `died_28d` (binary)	Logistic regression, complete case and multiple imputation
Q5	Positive balance, `positive_balance_d3` (binary)	ICU length of stay in survivors, `icu_los_days` (continuous, right-skewed)	Wilcoxon rank-sum (Mann-Whitney U) test, then linear regression on the log scale

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Exercise 8: Proposal builder 5 (what a good answer looks like)

The instructor fills this in for the running example, a trial of a restrictive fluid strategy against usual care.

8.1 Variables table

Role	Variable	How measured (unit, time point)	Type
Primary outcome	Death by day 28	Hospital record or phone call at day 28	Binary
Secondary outcome	Cumulative fluid balance at day 5	Fluid charts, ml/kg of admission weight, day 5	Continuous
Exposure / intervention	Restrictive vs usual fluid strategy	Randomised allocation	Binary
Confounder 1	Age	Years, at ICU admission	Continuous
Confounder 2	SOFA score	Points, first 24 hours	Ordinal score
Confounder 3	Serum lactate	mmol/L, at ICU admission	Continuous
Descriptive	Sex, main source of infection	At ICU admission	Binary, nominal

In a trial, age, SOFA and lactate are the key baseline variables for Table 1 (see the tip on the sheet).

8.2 Mini-CRF

#	Question exactly as the data collector will read it	Answer format	"Unknown / not done" option?
1	Study ID	FLU-[ ][ ][ ]	No
2	Age at ICU admission (years)	[ ][ ][ ]	□
3	Sex	□ male □ female	No

#	Question exactly as the data collector will read it	Answer format	"Unknown / not done" option?
4	Main source of infection	□ lung □ urine □ abdomen □ other	□
5	SOFA score in the first 24 hours (0 to 24)	[ ][ ]	□
6	Serum lactate at ICU admission (mmol/L)	[ ][ ].[ ]	□
7	Cumulative fluid balance from ICU admission to day 5 (ml)	□ + □ − [ ][ ][ ][ ][ ] ml	□
8	Alive at day 28	□ yes □ no	□

8.3 Data dictionary and data-quality plan

Variable name	Label (meaning)	Type	Unit / codes	Allowed values
fb_d5	Cumulative fluid balance, day 5	Continuous	ml/kg	-100 to 300
age	Age at ICU admission	Continuous	years	18 to 110
sex	Sex	Binary	1 = male, 2 = female	1, 2
died28	Died by day 28	Binary	0 = alive, 1 = died	0, 1

Data-quality plan: the study nurse checks every form for missing items each week. Each month the data manager runs range checks (for example age 18 to 110) and one consistency check (date of death not before ICU admission).

8.4 Data-management plan

Question	Model answer
Paper CRF with double data entry, or electronic data capture?	Electronic data capture (REDCap), with range and logic checks at entry
Coded study IDs? Where is the linking log kept, and who can open it?	Yes. The linking log is a separate, locked file. Only the principal investigator and the data manager can open it.
Validation rules at entry (two examples from your data dictionary)	Age 18 to 110 years. fb_d5 between -100 and 300 ml/kg.
Audit trail / cleaning log: who records changes?	REDCap records every change. The data manager keeps the cleaning log.
Backups (3-2-1) and data-sharing statement in the consent form	Encrypted backups: three copies on two kinds of media, one off site. The consent form says that coded data may be shared with other researchers.

Instructor note: quick check while modelling, and when reading drafts sent by email.

Element	Good enough	Needs work
Variables table	Outcome, exposure and ≥ 3 confounders, all with a type	Confounders that are actually outcomes, no types
Mini-CRF	6–10 items, units, time points, unknown option	> 15 items, free text, no units
Data dictionary	3–5 rows with codes and allowed ranges	Names with spaces, no codes

Element	Good enough	Needs work
Data management	EDC or double entry, coded IDs, audit trail, backups	Patient names in the data file, no backups